

Unsupervised Learning - Lab 04 - K-Means

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Agenda

- Motivation:
 - This fourth lab session aims to apply k-means algorithm and its variants to cluster an unknown matrix of data (with low dimensionality and continuous attributes).
- Steps:
 - Run the basic version of k-means with ONE random seed initialization **[TASK 1]**
 - Run multiple runs of the basic k-means **[TASK 2]**
 - Apply k-means++ **[TASK 3]**
 - Pre-process and then cluster **[TASK 4-5]**
 - Clustering for feature selection **[TASK 6]**

Instructions

- You are going to re-use already known packages (`matplotlib`, `seaborn`, `sklearn.preprocessing...`). Check the three previous lab solutions.
- Moreover, the `sklearn.cluster.Kmeans` package will be introduced to cluster data (see documentation [here](#)).
- Read the step-by-step instructions below carefully and write your own code to fill the missing steps in the Colab notebook (instructions are also reported in the notebook).

- numpy, pandas, scipy, matplotlib, seaborn, sklearn, `sklearn.cluster` (NEW!)

Check (and eventually re-use) those functions defined in previous solutions to compute centroids and find inter-/intra-cluster distances.